MSME IDEA HACKATHON 4.0

Idea Proposal

I. DETAILS OF THE IDEA

a. Title of the proposed Idea/Innovation:

DEVELOPMENT OF A PYROFIGHTER ROBOT FOR FIRE EXTINGUISHMENT IN URBAN AREAS

b. Team details:

Name	*Reg. No., Department/Udyam No.*
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c. Theme (select one):
Sustainable Development

d. Idea Sector (select one):
Miscellaneous Sector (Environment, Forests, Water & Sanitation; Foods, Beverages, FMCG, Consumer Goods; Infrastructure, Construction, Housing; IT, ITES, Electronics, White Goods, Telecommunication; Metals, Engineering, Machinery, Automation and Transportation, Automotive, E Vehicles, Railways, Aviation, UAV and any other sub-sector)

e. Summary of the idea:

A firefighting robot is an autonomous or semi-autonomous machine that detects and extinguishes flames in metropolitan settings. It can navigate complex situations, such as high-rise structures and confined spaces, which are challenging for human firefighters. Equipped with sensors, cameras, and firefighting gear, the robot aims to reduce response time, enhance safety, and minimize damage to property and lives.

f. Concept and Objectives:

Concepts:

Extinguishing robots are designed to be autonomous or semi-autonomous devices capable of detecting and extinguishing flames in populated areas. They can maneuver through tight spaces, narrow alleys, and high-rise buildings—locations that may be hazardous or difficult for people to access. The robots perform well in hot, smoke-filled environments, thanks to their sensors, cameras, and fire suppression equipment.

Objectives:

Quick Fire Detection: Use sensors and cameras to swiftly detect and identify fires.
Enhanced Safety: Reduce risks to firefighters by allowing robots to enter hazardous areas.
Effective Navigation: Utilize Wi-Fi modules to autonomously navigate through complex metropolitan structures.
Efficient Firefighting: Use water to extinguish various types of fires.
Reduce Property Damage: Extinguish fires quickly to minimize loss and damage..

g. Novelty/Uniqueness of the innovation:

The firefighting robot is a unique solution for urban fire situations due to its specialized capabilities and cutting-edge technology.

Autonomous Navigation in Complex Environments: Using sensors, the robot can independently navigate through challenging urban settings such as high-rise buildings, narrow alleys, and confined spaces—capabilities beyond those of human firefighters. This distinguishes it from standard firefighting tools.

Advanced Fire Detection and Targeting: Equipped with flame and ultrasonic sensors, the robot can analyze and identify heat sources and locate fire sites, even in low-visibility situations, responding to emergencies more quickly and accurately than traditional methods.

Reduced Risk for Human Firefighters: By operating in hazardous environments, such as areas with extreme temperatures or toxic smoke, the robot minimizes the exposure time of human firefighters to potentially fatal conditions, enhancing overall safety.

h. Features of the solution:

- **Autonomous Navigation**: The robot can navigate through complex urban environments, such as high-rise buildings, confined spaces, and narrow alleys, using sensors like cameras and ultrasonic sensors.

- **Advanced Fire Detection**: The robot's ultrasonic and flame sensors allow it to detect heat sources and flames even in low-visibility or smoke-filled environments, enabling early detection of fire threats.

- **Heat-Resistant Design**: With its robust construction and heat-resistant materials, the robot can operate effectively in high-temperature conditions, allowing it to work near active fires without sustaining damage.

i. Potential areas of application in the industry/market:

- **Industrial Facilities**: Handling combustible materials is common in factories and warehouses. In hazardous situations, firefighting robots can be used to extinguish fires, reducing damage and enhancing worker safety.

- **Public Events and Crowded Areas**: Robots can be employed for fire detection and suppression at large public gatherings, such as concerts or sporting events, providing a rapid response to potential fire outbreaks in densely populated areas.

- **Research and Development**: Research centers studying fire behavior, environmental monitoring, and the development of new fire suppression techniques can benefit from the technologies used in firefighting robots.

- **Military Applications**: Firefighting robots can assist military personnel and protect equipment by managing fire threats in combat zones or military installations.

j. Market data for potential idea/innovation:

- **Investments in Smart Technologies**: Investments in the Internet of Things (IoT) and robotics are predicted to surpass $2 trillion by 2025, driven by the growth of smart cities. This trend encourages the integration of robotic firefighting equipment into the infrastructure of smart cities.

- **Increasing Insurance Costs**: As fire incidents become more frequent, insurance premiums for businesses and property owners have increased. Robotic firefighting technologies are an attractive investment because they can help reduce fire-related losses.

- **Government Support and Regulations**: Governments are placing greater emphasis on fire safety by providing funding and establishing regulations for advanced firefighting technologies, which will increase the market potential for firefighting robots.

k. Current development status of the idea:

Prototypes of fire-fighting robots are being tested and improved, and the development process is progressing quickly. Technological advancements, collaborations, and increasing corporate interest point to a bright future for incorporating these robots into urban firefighting tactics. However, before a broad rollout can take place, further testing and improvements are required.

l. Expected time of completion:

Total Time Projected: 11–19 months

m. Risk factors and strategies to mitigate:

Technical Errors:

Hazard: Parts (such as sensors and mobility systems) may not work properly.
Damage Reduction:

- **Strict Testing:** Implement comprehensive testing procedures at every level of development.

- **Redundancy:** Design vital systems with backup components to ensure functionality.

Environmental Difficulties:

Hazard: Unpredictable urban settings (such as smoke and debris) may pose challenges for robots.
Damage Reduction:

- **Sturdy Navigation Systems:** Employ cutting-edge AI and sensor fusion for efficient navigation and obstacle identification.
- **Field Testing:** Conduct testing in diverse urban environments to adjust the design based on actual conditions.

Safety Issues:

Risk: Potential injury to individuals or property during operations.
Damage Reduction:

- **Security Protocols:** Clearly define emergency plans and operational rules.
- **Unsafe Mechanisms:** Include emergency stop buttons and automated shutdown features.

Background:

i. Introduction:

There are significant threats to life, property, and the environment due to the increased frequency and intensity of fires in urban settings. Although traditional firefighting techniques can be effective, they often face challenges from rapidly shifting conditions, limited resources, and difficult access. Firefighting robots offer an innovative solution by utilizing cutting-edge technologies such as robotics, artificial intelligence, and remote sensing. These robots are designed to autonomously detect, navigate, and extinguish fires, thereby increasing the effectiveness of firefighting operations while reducing risks to human firefighters.

ii. Problem(s) being addressed:

The firefighting robot tackles several important problems:

Quick Reaction:
Since fires can spread quickly, prompt responses are required. There may be delays for human responders, particularly in densely populated urban areas.

Safety Risks:
Smoke and structural hazards are common dangerous environments in which firefighters operate. Robots can navigate these hazardous situations without endangering human life.

Access to Difficult Areas:
Urban structures can be intricate, making it challenging for crews to reach every spot. Robots can effectively extinguish fires by navigating through confined locations and obstacles.

Resource Optimization:
Robotic assistance can help fire departments with limited manpower and equipment by freeing up human teams to concentrate on vital duties and allocate resources as efficiently as possible.

iii. Benefactor(s) of the solution:

- **Fire Departments:** Improved incident response will be possible with increased capabilities and efficiency.

- **Urban Residents:** Increased safety and reduced fire damage will help safeguard lives and property.

- **Insurance Companies:** Fewer fire-related claims and damage may lead to better risk assessments and lower payouts.

- **Local Governments:** Municipal budgets can benefit from enhanced public safety by reducing expenses associated with emergency responses and urban reconstruction.

iv. Prior art/solution(s), if any:

Drones for Fighting Fires:
Some fire departments employ drones equipped with thermal cameras to assess and detect fires. While drones are excellent for reconnaissance, they typically cannot extinguish fires.

Autonomous Firefighting Robots:
Companies and academic organizations have developed ground-based robots capable of independently detecting and extinguishing flames. These robots are known as autonomous firefighting robots. For example, the "Firefighting Robot" from the University of Maryland can navigate and extinguish small fires using foam or water.

Current Firefighting Technology:
Mobile water delivery systems and other cutting-edge technologies are used in conjunction with conventional firefighting tools, such as hoses and extinguishers. The effectiveness of robotic systems can be further enhanced by incorporating these technologies.

II. METHODOLOGY

a. How will the product be built and used?

Flame Sensors:
Flame sensors provide information to the system that helps locate the fire by detecting the presence of flames.

Raspberry Pi AI Camera (equipped with a Sony IMX500 sensor):
This camera gives the robot vision capabilities, allowing it to identify objects, locate fires, and analyze images using AI models.

Primary Controller (Raspberry Pi Zero 2W / Raspberry Pi):
This component processes sensor data, directs the robot's motion, and extinguishes fires.

Apple M2 Chip (8-core CPU, 10-core GPU) and 2023 iMac (8-core CPU, 8-core GPU):
Rather than being integrated directly into the robot, these are likely used for more complex AI training or image analysis activities during the development phase.

L298P Motor Driver:
This component regulates the BO motors that drive the wheels, allowing the robot to travel toward the fire source. The robot is mobile thanks to its four BO motors and wheels.

Mini Servo Motor:
This type of motor can be used to precisely control movements, such as adjusting the angle of the water pump nozzle.

Water Pump (12V):
Powered by the Raspberry Pi, this pump uses water to extinguish flames.

Power Management Components:
Three components manage power to various parts and operate the water pump: transistors, resistors (1K), and capacitors (104µF).

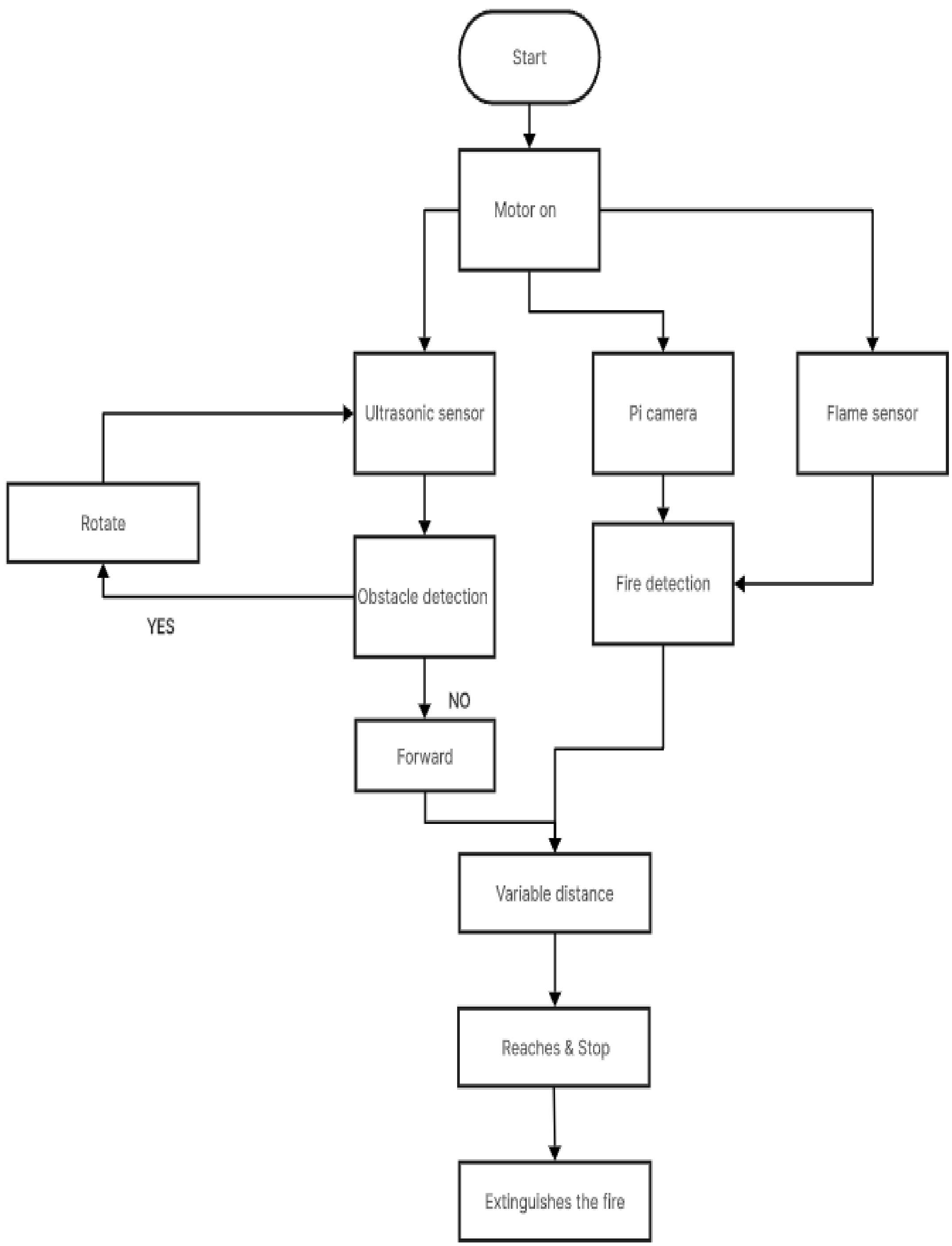
Lithium-Ion Battery:
This battery provides the necessary voltage and current for the Raspberry Pi, motor drivers, and sensors, powering the robot.

Jumper Wires and Breadboard:
These are essential for connecting various components and prototyping.

Apple Magic Keyboard & Mouse with Mouse Pad:
These are likely used for interfacing with the Raspberry Pi or the iMac during development and debugging.

Chassis:
The chassis serves as the structural framework of the robot, providing support and housing for all components, including the motors, sensors, and control systems.

b. Technical Details with Block Diagram/Flow Chart/Circuit Diagram etc.:



III. FINANCIAL REQUIREMENT

a. Total cost for implementation:

S.NO.	REQUIREMENT(DEVICE)	QUANTITY	COST PER DEVICE(Rs)	TOTAL COST (Rs)
1.	FLAME SENSORS	20	200	4000
2.	RASPBERRY PI AI CAM - BUILT WITH SONY IMX500 SENSOR	2	6,700	13,400
3.	CHACHIES	2	1000	2000
4.	RASPBERRY PI ZERO 2W	2	5,000	10,000
5.	L298P MOTOR DRIVER	2	1000	2000
6.	RASPBERRY PI	4	20,000	80,000
7.	APPLE M2 CHIP WITH 8-CORE CPU AND 10-CORE GPU	1	52,000	52,000
8.	APPLE 2023 iMAC 8-CORE CPU AND 10-CORE GPU	1	1,75,000	1,75,000
9.	APPLE MAGIC KEYBOARD	1	9,000	9,000
10.	APPLE MAGIC MOUSE	1	7,000	7,000
11.	MOUSE PAD	1	500	500
12.	BO MOTORS X 4+WHEELWITH SUSPENSION	2	50,000	1,00,000
13.	BREADBOARD	3	200	600
14.	MINI SERVO MOTOR	2	200	400
15.	WATER PUMP 12V	2	500	1000
16.	LITHIUM ION BATTERY	10	100	1000
17.	JUMPER WIRES	120		200
18.	TRANSISTOR	2	100	200

19.	CAPACITOR(104 uf)	10	2	20
20.	RESISTOR(1K)	10	2	20
21.	CHASSIS BODY	1	40,000	40,000

b. Details of the cost:

Sl. No.	*Item*	*Cost*
Technology related expenditure		
	IOT (IFTTT)	12,000
	CLOUD	1,04,000
	PUSH BULLET	10,000
	Other than technology expenditure	4,98,340
Other Expenses (Travelling expenses, consultation charges etc.)		
	Travel cost	80,000
	Food expenses	40,000
	3D Printing/Custom Enclosures	10,000
	Field Testing Costs	5,000
	MicroSD Card for Raspberry Pi: For data storage, logging, and software	2,000
	Robot Maintenance(e**g:** batteries, sensors**)**	9,000
	Soldering	5,000
	Shipping and Import Fees for Components	5,000
	Total:	7,80,340

REFERENCES

[1] P. Anantha Raj and M. Srivani, "Internet of Robotic Things Based Autonomous Fire Fighting Mobile Robot," *2018 IEEE International Conference on Computational Intelligence and Computing Research (ICCIC)*, Madurai, India, 2018, pp. 1-4, doi: 10.1109/ICCIC.2018.8782369.

[2] Shuo Zhang, Jiantao Yao, Ruochao Wang, Zisheng Liu, Chenhao Ma, Yingbin Wang, Yongsheng Zhao, Design of intelligent fire-fighting robot based on multi-sensor fusion and experimental study on fire scene patrol,

Robotics and Autonomous Systems,Volume 154,2022,104122,ISSN 0921-8890,
https://doi.org/10.1016/j.robot.2022.104122.

[3] Surabhi Srivastava a, Ritesh Yadav a,1,*, Usha Chauhan a Intelligent Robotic System For Fire Fighting , International Journal of Data Science Vol. 2, No. 2, December 2021, pp. 85-91, ISSN 2722-2039.